

Self-Supervised Network Reconstruction of Daily GNSS Time Series in California: A Benchmark for Signal-Preserving Common-Mode Removal and Transient Detection

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Abstract

Daily GNSS position time series resolve crustal deformation at the millimetre level, but noise that is correlated across stations limits their use for detecting aseismic transients. This common-mode noise is comparable in amplitude to the signals of interest, and it is spatially coherent, as real transients are. Classical common-mode filters and recent deep-learning denoisers both reduce it, yet they have been judged mainly by how much residual scatter they remove, a metric that also rewards the removal of genuine signal. Where signal recovery has been tested, it has not been examined as a function of spatial scale or compared with linear filters under a common protocol, and the multi-station deep-learning denoisers have been developed almost exclusively for subduction zones. Here we propose a self-supervised benchmark on 1,251 continuous stations in and around California processed by the Nevada Geodetic Laboratory. We withhold each station and predict it from its neighbours, so that the prediction captures the network-coherent part of the signal and the residual retains what is specific to the station. Because transients are themselves coherent, we also withhold all neighbours within an exclusion radius and vary this radius systematically, which measures the trade-off between noise suppression and signal leakage directly. We compare a hierarchy of models, from distance-weighted stacking and robust neighbour medians to ridge regression, frequency-dependent filtering and a graph-attention network, under identical temporal and spatial hold-out protocols. We score each model on reconstruction error, on the recovery of synthetic transients injected into real noise, and on the preservation of the 2019 Ridgecrest postseismic signal. The study will establish whether, and under which conditions, deep networks improve on linear spatial filtering, and it will provide an open benchmark and a calibrated noise model for near-real-time transient detection.

1 Introduction

Continuous GNSS networks record the positions of thousands of geodetic monuments every day, with formal uncertainties of about 1 mm in the horizontal and 3–5 mm in the vertical. The Nevada Geodetic Laboratory (NGL) processes more than 17,000 such stations worldwide with a single, uniform strategy and distributes the time series openly (Blewitt et al., 2018; Nevada Geodetic Laboratory, 2026). These records contain steady interseismic motion, coseismic offsets, postseismic relaxation, slow slip, magmatic inflation and hydrological loading, together with artefacts from antenna and receiver changes. Separating these contributions is the central task of geodetic time-series analysis.

The main obstacle is noise that is shared across stations. Errors in satellite orbits and clocks, in the realisation of the reference frame and in large-scale environmental loading produce

a common-mode component of several millimetres that is coherent over hundreds to thousands of kilometres (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). Because this component is spatially coherent, it is easily mistaken for a tectonic transient of similar wavelength. The debate over a reported precursory acceleration in stacked high-rate GNSS data before large earthquakes shows what is at stake (Bletery and Nocquet, 2023). The signal disappears once a common-mode correction is applied, and the disagreement that followed is precisely about whether such corrections remove only noise or also real deformation (Bradley and Hubbard, 2023; Bletery and Nocquet, 2025).

Our preliminary analyses point to the same problem. We examined departures from a fitted trajectory model at stations around the East Anatolian Fault between the 2020 Elazığ and 2023 Kahramanmaraş earthquakes, and at eleven continuous stations within 80 km of the 2019 Ridgecrest sequence. In Türkiye the public record has gaps from 2016 to 2020 and from 2021 to 2022, which rules out a continuous analysis of the interval between the two earthquakes. At Ridgecrest the record is continuous. Coseismic offsets reach 0.63 m in the near field, and 10–25 mm of postseismic displacement accumulates over the following two years. Before the mainshock, by contrast, east departures stay below 1 mm, while north departures shift by 0.3–1.7 mm with the same sign at every station, regardless of distance from the rupture. A coherent vertical depression of 3–6 mm in early 2019 coincides with an exceptionally wet winter and later recovers. Every apparent anomaly we found is explained by network-wide noise, hydrological loading or equipment changes rather than by deformation on the fault. We therefore turn from the search for precursors to the problem that precedes it: separating coherent noise from coherent signal.

A natural way to make this separation is to predict each station from its neighbours. The prediction can only reproduce what the station shares with the surrounding network, so monument noise, local site effects and station-specific artefacts remain in the residual. Whether the prediction or the residual is the useful product depends on the spatial scale of the noise relative to that of the signal, which is why such methods must be tested explicitly for signal preservation. The same principle is familiar from seismology. Tilt and compliance noise is removed from ocean-bottom seismometer records by predicting the vertical channel from the horizontal and pressure channels (Crawford and Webb, 2000; Janiszewski et al., 2019), and distributed acoustic sensing records are denoised by predicting each channel from its neighbours (Batson and Royer, 2019; van den Ende et al., 2023). Graph neural networks for spatiotemporal kriging and imputation are trained in the same way, reconstructing randomly withheld sensors from their neighbours (Wu et al., 2021; Cini et al., 2022; Marisca et al., 2022). To date, this approach has not been examined systematically for dense geodetic networks.

Here we test the hypothesis that a deep network improves on linear spatial prediction for GNSS common-mode removal only where the relation between stations is nonlinear, and only if it does not absorb real transients in the process. The study has three objectives. First, we build an analysis-ready, leakage-controlled data set of daily displacements for the California network, with fixed temporal and spatial splits. Second, we define a self-supervised reconstruction benchmark and evaluate a hierarchy of linear and nonlinear models on it, with signal preservation as a primary criterion. Third, we determine whether a deep network gives a measurable improvement over linear filtering and, if so, in which noise and signal regimes. The resulting model and its calibrated residuals will form the basis of a continual, near-real-time transient detector.

2 Literature Review

2.1 Common-mode error and network filtering

Network filtering of GNSS data began with the Southern California Integrated GPS Network, where the common-mode error was estimated as the weighted mean of the residuals at the other stations and subtracted from each series (Wdowinski et al., 1997). Principal component anal-

ysis and the Karhunen–Loève expansion extended this to common modes that vary in space (Dong et al., 2006), and variational Bayesian independent component analysis provided a blind source separation framework that tolerates missing data (Gualandi et al., 2016). At the level of individual stations, robust trend estimation (Blewitt et al., 2016) and greedy automatic trajectory decomposition (Bedford and Bevis, 2018) address velocity estimation and the separation of steps, seasonal terms and transients. Correlation-weighted spatial filtering estimates each station’s common mode from the other stations, weighted by correlation and distance, and outperformed principal component filtering in the western United States (Tian and Shen, 2016; Kreemer and Blewitt, 2021). The closely related common-mode component imaging (Kreemer and Blewitt, 2021) takes a correlation-weighted median over the nearest stations, including the target station itself when it qualifies as a filter station. Applied to about 2,500 stations in western Europe, it reduced residual scatter by 39–50%. The authors attribute much of this gain to local common-mode transients being absorbed into the estimated common mode, and suggest recovering such transients by differencing common modes estimated at local and broad scales. Common-mode component imaging was evaluated by scatter reduction and velocity repeatability, without injected signals. Offset detection has been compared in a blind community exercise (Gazeaux et al., 2013), and transient detection in the SCEC validation exercise (Lohman and Murray, 2013). Neither exercise has been repeated with machine-learning methods.

2.2 Deep learning for geodetic denoising

Deep-learning approaches to GNSS denoising are recent and few. Mastella et al. (2025) trained a single-station model that couples a step detector with a bidirectional recurrent network, using synthetic series with noise generated by an adversarial network trained on about 5,000 NGL stations. On real stations, the model is scored against the residuals of a greedy trajectory decomposition rather than an independent reference, and its gains are confined to periods shorter than about 80 days. The authors name the inclusion of neighbouring stations as the next step. Network-aware methods have so far been developed for Cascadia. Costantino et al. (2024) proposed a spatiotemporal graph neural network for extracting slow slip signals and evaluated it on the recovery of synthetic events and on agreement with tremor activity. The same group later inverted fifteen years of denoised data for the slip history of the subduction interface (Costantino et al., 2026). Bachelot et al. (2025) compared a graph neural network with stack filtering on two Cascadia data sets and reported horizontal common-mode reductions of more than 70% and 30%, while noting that stack filtering may still be preferable for some applications. Most recently, Teutschmann et al. (2026) pretrained a self-supervised foundation model on 17,651 NGL stations. Their model treats stations independently, and the authors list graph-based spatial encoders, denoising and imputation as future work. To our knowledge, no graph-based or masked-reconstruction model has been trained on the California network, and no deep-learning denoiser has been compared with common-mode component imaging; existing comparisons use simpler stacking filters (Mastella et al., 2025; Bachelot et al., 2025).

2.3 Transient detection with synthetic training data

Labelled geodetic transients are scarce, so the most successful detectors are trained on synthetic signals. Costantino et al. (2023) injected 60,000 synthetic slow slip events, computed with elastic dislocation models (Okada, 1992), into surrogate noise derived from real Cascadia series, and trained a convolutional, attention-based detector on 135 stations. The detector recovered 35 of 40 events in an independent catalogue (Michel et al., 2019) and reached a detection threshold of about M_w 6.2–6.4. The authors caution that regions with frequent seismicity and overlapping postseismic relaxation would make the problem harder, and that the approach is region-specific. Other strategies include training on real labelled windows, which reached about 75% accuracy in southwest Japan (Tanaka et al., 2025), and convolutional autoencoders for InSAR time series

(Rouet-Leduc et al., 2021). Classical network-based transient detectors include the network strain filter (Ohtani et al., 2010), Kalman smoothing with principal component analysis (Ji and Herring, 2013) and sparse estimation (Riel et al., 2014); these are the natural classical baselines for the detection stage of this work. In California, the only systematic statewide analysis of transient deformation is the classical study of Klein et al. (2019), which used about 1,000 stations over 1999–2018 and separated postseismic, hydrological, anthropogenic and magmatic signals. Across this literature, false-alarm rates measured on real data are rarely reported.

2.4 Precursory deformation and forecasting

The geodetic evidence for precursory deformation remains contested. The hour-scale acceleration reported by Bletery and Nocquet (2023) from 3,026 high-rate series before 90 earthquakes of $M_w \geq 7$ vanishes after common-mode correction (Bradley and Hubbard, 2023). The original authors accept this but argue that the correction removes tectonic signal (Bletery and Nocquet, 2025). An independent study of stacked tiltmeter records before the 2011 Tohoku-oki earthquake found no precursory signal above a detection limit of about M_w 6.4 (Hirose et al., 2024), and the months-long, thousand-kilometre-scale motions reported before great subduction earthquakes (Bedford et al., 2020) have also been questioned. The best-supported cases are local and slow, such as the eight-month slow slip event before the 2014 Iquique earthquake (Socquet et al., 2017). None of them has been shown to be predictive, that is, to occur more often before large earthquakes than at other times.

In earthquake forecasting, geodesy has improved skill mainly through static strain-rate maps combined with smoothed seismicity, as shown in prospective global tests (Strader et al., 2018; Bayona et al., 2020). Time-dependent geodetic inputs have not been tested prospectively against the epidemic-type aftershock sequence (ETAS) model, and relative afterslip moment does not correlate with aftershock productivity across 41 earthquakes (Churchill et al., 2022). In California, neural forecasting models trained on catalogues alone have not outperformed ETAS (Stockman et al., 2024), and simple models often match complex networks (Mignan and Broccardo, 2019, 2020). Postseismic deformation has been forecast with data-driven methods for individual subduction earthquakes (Yamaga and Mitsui, 2019; Kano et al., 2020), but not across several events or in a strike-slip setting. Likelihood-based tools for evaluating forecasts are available in pyCSEP (Savran et al., 2022).

2.5 Synthesis

Three gaps emerge from this literature. First, network-aware deep-learning denoising has been developed and evaluated only in subduction settings. Second, denoising methods are compared mainly by scatter reduction. The exceptions test the recovery of synthetic signals (Mastella et al., 2025; Costantino et al., 2024) or agreement with tremor (Costantino et al., 2024), but none examines how recovery depends on the spatial scale of the transient relative to that of the filter; Costantino et al. (2024) analyse recovery error only as a function of signal-to-noise ratio. This matters for the science built on denoised data: in Cascadia, inferred slow-slip scaling laws, b-values and stress drops all change with the detection threshold applied after denoising (Costantino et al., 2026). Third, no study has determined whether nonlinear models improve on linear spatial prediction when both are trained and tested under identical, leakage-controlled conditions. Closing these gaps is a prerequisite for transient detection, time-dependent forecasting and precursor testing alike, because each depends on knowing which signals survive noise removal.

3 Proposed Method

3.1 Data

We use NGL daily IGS20 solutions for all stations between 32.4° and 42.1°N and between 124.6° and 114.0°W with at least 1,095 daily solutions, which gives 1,381 stations in California and adjacent parts of Nevada, Oregon and Baja California (Nevada Geodetic Laboratory, 2026). We restrict the analysis to 2008 onward, when network density is approximately constant. We remove 130 stations. Thirty have less than one year of data after 2008. Twenty-two show non-tectonic site motion, such as groundwater-driven subsidence in the Central Valley, with a robust residual scale above 10 mm in the horizontal or 30 mm in the vertical. Seventy-eight are co-located duplicates within 1 km of a longer-running station. The remaining 1,251 stations provide 6.45×10^6 station-days, with 827 stations operating in mid-2008 and 1,010–1,025 between 2019 and 2022. The median completeness within each station’s operating period is 96%. In 2018, the median distance to the nearest station was 9.7 km, and a typical station had 23 neighbours within 50 km and 75 within 100 km.

3.2 Preprocessing and data partitioning

For each station i and component $c \in \{E, N, U\}$, we fit a trajectory model with a constant, a linear rate, annual and semi-annual harmonics and Heaviside steps at catalogued epochs by least squares,

$$x_{i,c}(t) = a + v(t - t_0) + \sum_{k=1}^2 [s_k \sin(2\pi kt) + c_k \cos(2\pi kt)] + \sum_j b_j H(t - t_j) + r_{i,c}(t). \quad (1)$$

We fit the model on the training period only, so that no information from the validation or test periods enters the inputs. We subtract the estimated steps at equipment changes but keep earthquake offsets in the residuals, because offsets are spatially coherent and should be predictable from the network. Observations more than five robust standard deviations from a running median are treated as missing. The residuals $r_{i,c}(t)$, in millimetres, together with the formal uncertainties $\sigma_{i,c}(t)$ and an availability mask, form an array of stations by days by components. For each station and each exclusion radius R (Section 3.4), the sixteen nearest stations at distances $R < d \leq 300$ km form its neighbourhood graph.

We assign 2008–2019 to training, 2020–2021 to validation and 2022–2026 to testing, and evaluate the test period once. We withhold 173 stations entirely, chosen to be spread across the network, to test generalisation to unseen sites. For all stations within 150 km of the Ridgecrest rupture, we exclude the interval from July 2019 to June 2020 from training, so that every model can be tested on postseismic data it has never seen.

3.3 Self-supervised reconstruction task

Let $\mathcal{M}$ be a set of withheld station-day-component entries. A model f_θ receives the array with the entries in $\mathcal{M}$ removed and predicts them as $\hat{x}_{i,c}(t)$. We fit models with trainable parameters by minimising the Gaussian negative log-likelihood over the withheld entries,

$$\mathcal{L}(\theta) = \sum_{(i,t,c) \in \mathcal{M}} \left[\frac{(r_{i,c}(t) - \hat{x}_{i,c}(t))^2}{2\sigma_{i,c}^2(t)} + \log \sigma_{i,c}(t) \right], \quad (2)$$

so that the formal uncertainties down-weight unreliable observations. We use three masking patterns: randomly scattered entries covering 10% of the data, contiguous 30-day blocks at individual stations, and complete withholding of the held-out stations. The test-period masks are generated once with a fixed seed and shared by all models. We interpret the reconstruction

$\hat{x}$ as the network-coherent component and the residual $r - \hat{x}$ as the station-specific component. Cleaned series and all signal-preservation scores are computed leave-one-out, with the target station withheld for the entire period. Otherwise a flexible model could simply reproduce the target and produce a trivially small residual.

3.4 Exclusion radius and temporal context

Leave-one-out prediction removes whatever is coherent at the scale of the station spacing, and postseismic relaxation, slow slip and hydrological loading are all coherent at that scale. The common-mode error, by contrast, is coherent over hundreds of kilometres (Kreemer and Blewitt, 2021), whereas a transient with footprint L reaches stations only to distances of order L . We therefore withhold all neighbours closer than an exclusion radius R and treat R as an experimental variable rather than a tuning parameter, with $R \in \{0, 10, 25, 50, 100\}$ km. The same device is used in buffered spatial cross-validation, where observations within a buffer around each held-out point are excluded so that spatial autocorrelation cannot inflate apparent skill (Le Rest et al., 2014). It also generalises the suggestion of Kreemer and Blewitt (2021) to recover local transients by differencing common modes estimated at local and broad scales. We fit and evaluate every model separately at each radius. Larger radii should preserve more transient signal at the cost of weaker noise suppression. They also emulate sparser networks: at $R = 50$ – 100 km, the California network approaches the station spacing of the public network in Türkiye, where the median distance to the nearest station is about 60 km.

A second route by which signal can leak into the prediction is the target’s own history. A model that sees the target on days adjacent to a withheld interval can extrapolate the onset of a transient into it. We therefore distinguish three temporal contexts: same-day prediction, which uses neighbours on the same day only; causal prediction, which uses data up to the prediction day; and two-sided prediction, which uses a window on both sides. Use of the target’s own unwithheld data is a separate factor. We disable it for cleaned series and signal-preservation tests, and allow it only for gap-filling scores, where results state it explicitly. Models M1–M3 are same-day by construction, M4 is two-sided, and we evaluate M5 in all three contexts with and without own history. The causal context is the one the continual detector of Section 3.7 requires.

3.5 Model hierarchy

We compare six models of increasing capacity, eight variants in total (Table 1). M0 predicts zero and sets the floor. M1 is the distance-weighted mean of neighbouring residuals, as in regional stacking (Wdowinski et al., 1997). M2 replaces the mean with a robust, outlier-resistant neighbour statistic, in the spirit of common-mode component imaging (Kreemer and Blewitt, 2021). M3 is a ridge regression on the sixteen neighbours, fitted on the training period. Missing neighbour values are set to zero and paired with availability indicators, and neighbours are randomly withheld during fitting, so that one model handles any gap pattern. Weights estimated for one station cannot be applied to a station absent from training, so M3 comes in two variants. M3a estimates a separate weight vector for each station and is evaluated on stations seen in training. M3b estimates one weight vector shared by all stations, with neighbour residuals interacted with inter-station distance and azimuth, and therefore also applies to the held-out stations. M4 extends M3 to frequency-dependent weights by regressing on band-pass filtered neighbour series, analogous to a multichannel Wiener filter, with the same per-station (M4a) and pooled (M4b) variants.

M5 is a graph-attention network. For a target station and its neighbours, a temporal convolutional network with weights shared across stations encodes a 64-day window of residuals, uncertainties and availability flags. The target embedding attends to the neighbour embeddings, with attention conditioned on inter-station distance and azimuth, and a decoder predicts the

withheld values. The network has about 5×10^5 parameters. We keep it this small on purpose: the common mode is a single realisation per day, so the effective number of independent training examples for the network-wide component is the number of training days, about 4,400 for 2008–2019, not the number of station-days. Training samples random target stations, windows and masks from the training period, and early stopping uses the validation period. We train a separate network for each combination of exclusion radius, temporal context and own-history setting that we evaluate. Training fits on a single consumer graphics processor with 8 GB of memory.

Table 1: Model hierarchy for the reconstruction benchmark.

Model	Description	Trainable	Reference approach
M0	Zero prediction	No	Floor
M1	Distance-weighted neighbour mean	No	Wdowinski et al. (1997)
M2	Robust neighbour statistic	No	Kreemer and Blewitt (2021)
M3a	Ridge regression on neighbours, per station	Yes	Linear spatial prediction
M3b	Ridge regression on neighbours, pooled over stations with geometric features	Yes	Linear spatial prediction
M4a	Band-split ridge regression, per station	Yes	Multichannel Wiener filter
M4b	Band-split ridge regression, pooled over stations	Yes	Multichannel Wiener filter
M5	Graph-attention temporal network	Yes	Bachelot et al. (2025); Costantino et al. (2024)

3.6 Evaluation

We score each station separately and summarise the scores by the median over stations, with the interquartile range, so that a few stations with large earthquake offsets cannot dominate the result. For each station, we split the error on the withheld entries into a slow part, its 61-day running median, and a fast part, the remainder. The fast part measures how well a model predicts day-to-day and week-to-week common-mode noise and is the primary score. The slow part contains trend drift accumulated since the training period, earthquake offsets and slow transients; we report it separately and judge it with the signal-preservation tests below. Both parts are expressed as root-mean-square errors normalised by the station’s noise level in the training period, and reported for each masking pattern, component and hold-out type. Entries within 30 days of an earthquake of magnitude 6 or larger that may have offset the station are left out of these scores. Stations that show site problems after the training period, either a more than threefold increase in day-to-day scatter or a jump that no logged equipment change or earthquake explains, are left out of scoring altogether. We adopted these rules after a first scoring run in which errors pooled over all stations were dominated by a few stations near the Ridgecrest rupture and by one station that was physically relocated in 2025. Because scatter reduction alone rewards the removal of real signal, we add two further criteria. First, we inject synthetic transients into the test-period data before reconstruction. These are spatially smooth displacement fields with Gaussian footprints of 10–100 km, amplitudes of 2–10 mm and durations of 10–90 days, followed in a second phase by elastic dislocation sources on strike-slip and creeping faults (Okada, 1992). The recovery ratio

$$\rho = \frac{\langle r - \hat{x}, s \rangle}{\langle s, s \rangle}, \quad (3)$$

where s is the injected signal, gives the fraction of the transient retained in the residual. We map it as a function of footprint, amplitude, duration and exclusion radius. For each model, plotting common-mode reduction against recovery ratio as R varies summarises the trade-off between noise suppression and signal leakage in a single curve. Second, we compute the fraction of the observed Ridgecrest postseismic displacement retained in the residual over the excluded interval. Velocity repeatability between disjoint time windows checks whether long-period signal is distorted. We assess differences between models with paired bootstrap resampling over days and stations, and report the spread across random seeds for the trainable models.

3.7 Extension to continual transient detection

If a reconstruction model preserves injected transients, its residuals become the input to a transient detector. We will calibrate a multi-scale cumulative-sum statistic on the residuals, combined with a spatial-coherence criterion that separates isolated station anomalies from clusters of neighbouring stations, to a set number of false alarms per station-year on quiet intervals. We will then run the detector in a strictly chronological replay of 2010–2026, updating the reconstruction model as new data arrive. To keep slow transients from being absorbed as normal behaviour, model updates will be restricted to windows not flagged as anomalous and regularised against catastrophic forgetting (Kirkpatrick et al., 2017). Injected events will measure detection delay as a function of transient size and duration, and the extent to which adaptation suppresses slow signals. This extension depends on the outcome of the benchmark; we describe it here to show where the research is heading.

4 Expected Findings

We design the study so that every plausible outcome is informative, and we test three hypotheses.

First, we expect linear spatial prediction to capture most of the common-mode noise. M3 should reduce the normalised reconstruction error well below M1 and M2, because station-specific neighbour weights can represent spatial variations of the common mode that a uniform stack cannot. If M2 performs as well as M3, the robust-median approach of common-mode component imaging would be confirmed as close to optimal for this network, which is itself a useful result.

Second, we expect a nonlinear model to improve reconstruction only where the relation between stations depends on frequency, on station availability or on transient regimes such as postseismic intervals. M5 should outperform M3 and M4 by a modest margin, with the largest gains at stations with intermittent neighbours and at periods near the draconitic and fortnightly frequencies of GNSS-specific noise. The decisive question is whether this gain comes with equal or better transient recovery. Lower reconstruction error combined with lower recovery ratios would mean the network is over-smoothing, absorbing real signal into the coherent component, and would argue against using it for transient detection.

Third, we expect transient recovery to depend mainly on the ratio of the transient footprint to the exclusion radius. A transient should be largely absorbed when its footprint L exceeds R , because it then reaches the stations used for prediction, and largely preserved when R exceeds L , with recovery rising from near zero towards one across the transition. Noise suppression should decline more slowly with R , because the common mode is coherent over larger distances than the transients we consider, so an intermediate radius should preserve transients of a given scale while keeping most of the noise reduction. We also expect own history to lower the recovery of transients with short onsets in the deep model, and comparing runs with and without it will quantify this temporal leakage. The resulting maps will show the spatial scale below which daily GNSS can resolve aseismic transients after common-mode removal, and will provide detection thresholds for later searches for slow deformation in California.

The main deliverables are an open, leakage-controlled benchmark for network reconstruction of daily GNSS data, including the data array, splits, masks and injected signals; a quantitative comparison of linear and deep models under identical conditions; and maps of transient recovery as a function of spatial and temporal scale. If the deep model fails to surpass linear prediction on transient recovery, we will report that with equal weight, since it would show that the added complexity is not warranted for this application. In either case, the calibrated residuals will support the continual detection framework outlined above and, later, tests of time-dependent geodetic covariates in likelihood-based earthquake forecast evaluation.

AI Usage Declaration

An AI assistant, Claude Code running the Claude Opus 5.5 model (Anthropic), was used to help design the benchmark, write the software and its tests, run the preliminary analyses and produce the figures, search and summarise the literature, and draft and edit this text. Bibliographic entries were generated from Crossref and arXiv metadata. The research question and the central idea of predicting withheld stations from their neighbours are the author’s. The author directed the work, reviewed all AI-generated material and is responsible for the content. Some cited papers have so far been assessed from their abstracts and will be read in full before any results are submitted for publication.

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